3
280
4
1120

Midterm Examination
ECO 101: Principles of Microeconomics
Department of Economics
Shiv Nadar University
Monsoon 2016.

This is a one and a half hour long closed book, closed notes in class examination. Answer all the questions, and be clear and precise in your answers. Any act of *academic dishonesty* will be taken very seriously. *All the best.*

Question 1: Incidence of Commodity Taxation (20 points)

Consider the following model for the toothbrush market in Delhi. Suppose the aggregate demand for brushes in Delhi is given by $Q_D = 900 - P/2$ where P denotes the price and Q denotes the quantity of brushes in terms of ~~thousands~~ of brushes demanded. The aggregate supply for brushes in Delhi is given by $Q_S = P/4$. $900 - \frac{P}{2} = \frac{P}{4}$

(a.) Compute the toothbrush market equilibrium. What are the equilibrium price and quantity? (4 points)

(b.) Now suppose a tax of $t = 60$ is imposed on each brush that is purchased. Write down the changed post-tax demand and supply equations and compute the brush market equilibrium with the tax. What are the equilibrium price that the sellers receive, and what is the post-tax equilibrium quantity transacted? (6 points)

$Q_D = 900 - \left(\frac{P}{2} + T\right)$

$\Rightarrow \frac{P}{4} = 900 - \frac{P}{2} - 60$

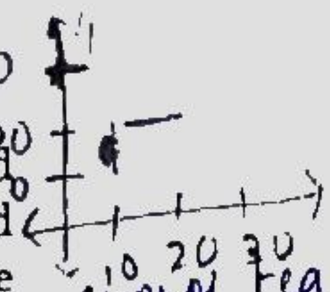
$\Rightarrow \frac{3P}{4} = \cancel{840}\ 280$

$P = 280 \times 4$

(c) What is the price paid by the buyers and therefore, what is the incidence of the tax? Explain the intuition for the key factors that determine the incidence. (5 points)

(d.) Compute and graphically depict dead-weight loss due to the tax. (5 points)

Question 2: Consumer Choice and Comparative Statics (30 points)



Consider an individual, Bruce, who considers Tea (X) and Sugar (Y) as *compliments* and therefore likes to consume it together. His preferences over Tea and Sugar are represented by: $u(x, y) = \min\{3x, 2y\}$. The price of a cup of Tea is INR 3, and that of a cup of Coffee is INR 2, and she has INR 60 to spend on her consumption of beverages.

(a) Draw the family of indifference curves that represents Bruce's preferences in a properly labelled diagram. From the utility function, can you specify how many *cubes* of sugar Bruce takes with his each cup of Tea. (3 points)

$4+6+4+5$

$3T = 2S$ 19 20

$\Rightarrow T = \frac{2}{3}S$

(b) Draw the budget line in a property labelled diagram, and interpret the intercepts and the slope of the budget line. (2 points)

(c) Find out the initial point of consumer equilibrium, i. e., how many cups of Tea and Coffee are consumed by Bruce? Is the usual consumer equilibrium condition of equality between *MRS* and *price ratio* satisfied at this equilibrium point? Using your finding, explain why that combination emerges as equilibrium? (5 points) why it is satisfied in a equilibrium

Now suppose the government impose a tax on Tea. The market demand and supply of tea are such that as a result, price of Tea happened to increase to INR 5 per cup.

$\min\{3x, 2y\} = 60$

(x, y)

$20, 30$

$\frac{1120}{4}$

$\frac{1120}{-8}$ 32

$\frac{1120}{-8}$ 3

$3x + 2y = 60$

$(20, 30)$

$900 - \frac{1120}{2} - 60$

$840 - 560$

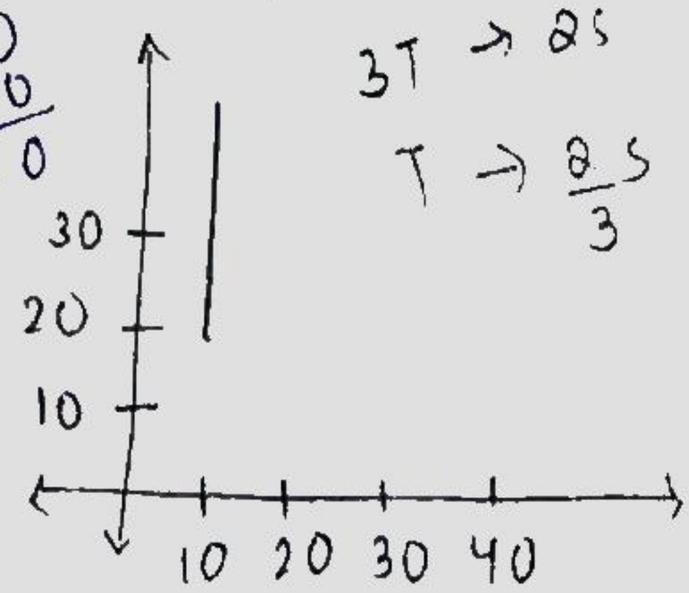
280

$\cancel{840}$ 560

$3x, 2y$

$30, 20$

$\{3x, 2y\} = 60$



$3T \to 2S$

$T \to \frac{2}{3}S$

(d) How many cups of Tea and Coffee will Bruce consume in this changed situation? (4 points)

(e) Decompose this price effect into income and substitution effect (algebraically). Show the decomposition in a properly labelled diagram and provide intuition for your answer.(6 points)

(f) In order to find out the substitution effect, the consumer is compensated in a way such that she can afford the initial consumption bundle at the changed price. After such a compensation here, is Bruce better off, worse off, or as well of as she was before price change? In general, what determines the direction of change in consumer satisfaction due to such a compensation? (4 points)

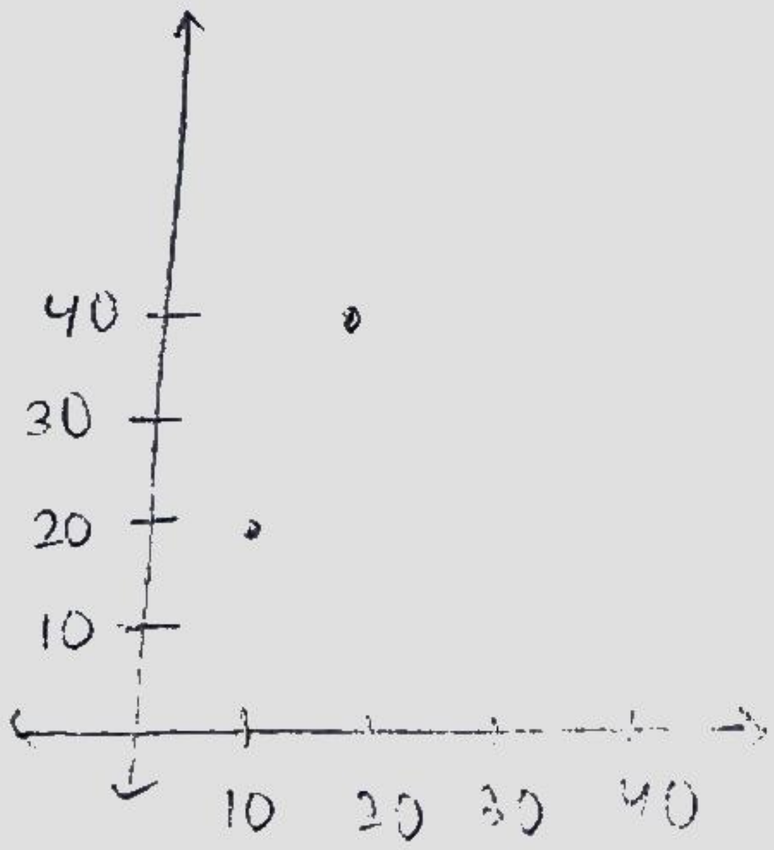
(g) Draw Bruce's demand for Tea as a function of price of Tea, keeping the price of Sugar unchanged at INR 2 per cup. Needless to say, all other factors that can influence the demand for Tea are also kept unchanged. Compute the demand as a function of all possible prices of Tea, and explain the shape of the demand curve. (6 points)

Note: The demand curve is the relationship between price and quantity demanded of a commodity, keeping everything else unchanged.

Hint: Use your answers of part (b) and (c) to see how the optimal choice (or, demand) changes with price, and then think about what the demand would be for all other possible prices.

Date: September 30, 2016.

Instructor: *Subhra K. Bhattacharya.*



(30, 20) 60 40

(3x, 2y)

(

40, 60